

Problem 10.6

Bruce deposits 100 into a bank account. His account is credited interest at a nominal rate of interest of 4% convertible semiannually. At the same time, Peter deposits 100 into a separate account. Peter's account is credited interest at a force of interest of δ . After 7.25 years, the value of each account is the same. Calculate δ .

Solution

Bruce's nominal interest rate is 4% convertible semiannually, so the effective interest rate per half-year is

$$\frac{0.04}{2} = 0.02.$$

There are

$$2(7.25) = 14.5$$

half-year periods in 7.25 years. Therefore, Bruce's account accumulates to

$$100(1.02)^{14.5}.$$

Peter's account accumulates under a constant force of interest to

$$100e^{7.25\delta}.$$

Since the values of the two accounts are equal,

$$100e^{7.25\delta} = 100(1.02)^{14.5}.$$

Canceling 100 gives

$$e^{7.25\delta} = (1.02)^{14.5}.$$

Taking natural logarithms,

$$7.25\delta = 14.5 \ln(1.02).$$

Thus,

$$\delta = \frac{14.5}{7.25} \ln(1.02) = 2 \ln(1.02).$$

Therefore,

$$\delta = 0.03960525.$$

Hence, the force of interest is

$$\boxed{\delta = 0.03960525}.$$